Module 0 Homework 4 Challenge

Provide the best answer for each of the following questions.

Time Remaining 00:59:12

Main Question Set

Solve six of the first ten questions to achieve a passing score of 60%. Don't miss the learning opportunity in the rest of the problems, though -- spend the full time and attempt every one of them!

1

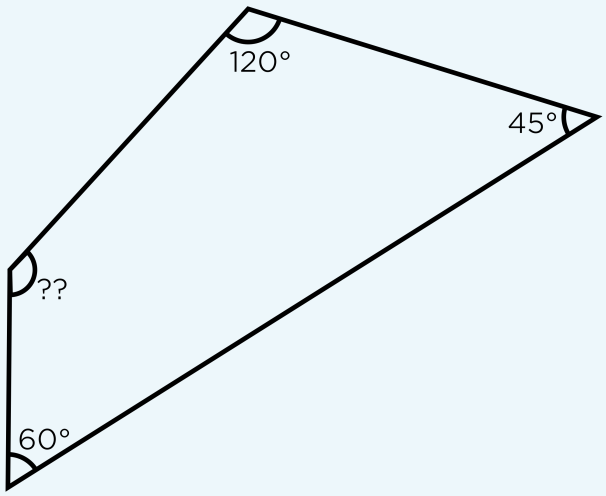
Review Later

Perfect square numbers have which possible last digits?

- $1, 2, 3, 4, 5, 6, 7, 8, 9, 0$
- $1, 3, 5, 7, 9$
- $2, 4, 6, 8, 0$
- $1, 4, 9, 6, 5, 0$
- 0

2

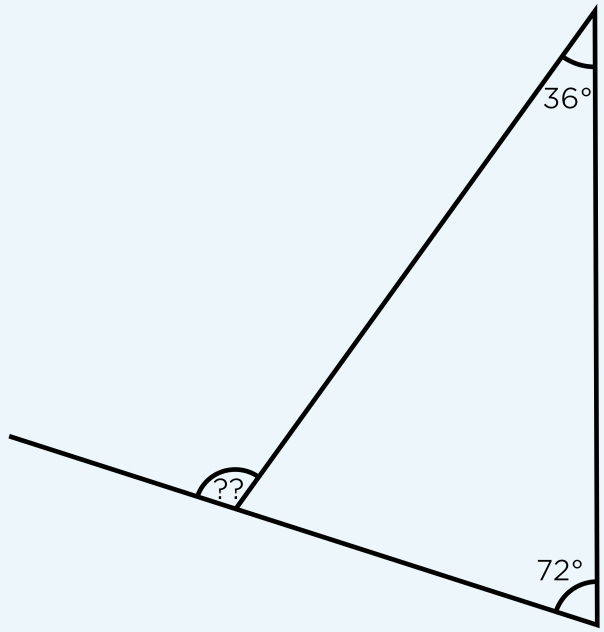
Review Later



How many degrees are in the marked angle?

3

Review Later



How many degrees are in the marked angle?

4

Review Later

What is $\left(100-29\right)+\left(382382-100\right)+\left(129-382382\right)$?

5

Review Later

The sum of the interior angles in any polygon is always a multiple of which of these numbers?

- 101
- 53
- 107
- 180

360

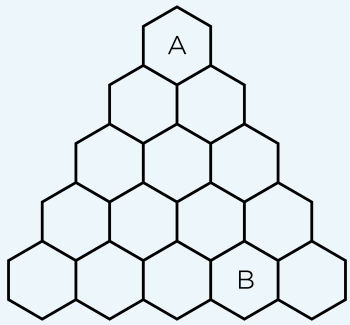
6

Review Later

What is $\frac{5}{14} - \frac{1}{2} + \frac{1}{7}$?

7

Review Later



How many different paths are there from A to B if you can only go from one hexagon to a neighboring hexagon that is one of the two below it?

- 4
- 1
- 33
- 37
- 2

8

Review Later

What are the last two digits of 201912^2?

9

Review Later

What is the last digit of 271828^6?

10

Review Later

What is $\frac{6}{17} \times \frac{7}{5} \times \frac{8}{3} \times \frac{5}{6} \times \frac{3}{7}$ expressed in simplest fractional form?

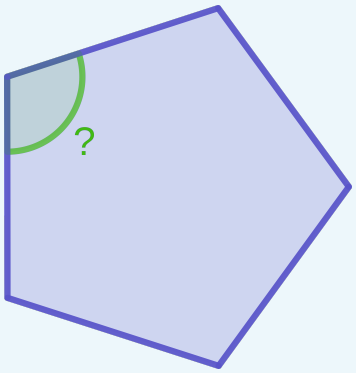
Your answer should be a fraction without spaces, e.g. 1/2.

Bonus Challenge Questions

These optional questions will not count towards your final score. However, we strongly suggest trying them for the benefit of the challenge!

11

Review Later



How many degrees are there in the interior angle of a regular pentagon?

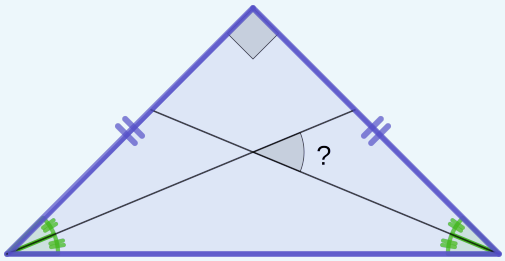
12

Review Later

What is the remainder when 13^3 is divided by 8?

13

Review Later



This is a special right triangle with two legs that are the same length. The two lines crossing through the triangle are *bisectors* (they split the angles they go through into half). How many degrees are in the marked angle?

14

Review Later

Positive integer a has remainder 1 when divided by 7, while positive integer b has remainder 1 when divided by 3.

What is the smallest possible value of b that makes $a \times b$ always a multiple of 7?

15

Review Later

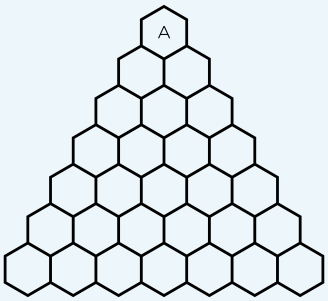
Which of the following is the same as the fraction shown below?

$$\frac{6\sqrt{6} - 5\sqrt{5}}{30\sqrt{30}}$$

- $\frac{1}{5\sqrt{5}} - \frac{1}{6\sqrt{6}}$
- $\frac{1\sqrt{30}}{30\sqrt{30}}$
- $\frac{6\sqrt{180} - 5\sqrt{150}}{30}$
- $\frac{1\sqrt{6}}{30\sqrt{30}}$
- $\frac{6}{30\sqrt{6}} - \frac{5}{30\sqrt{5}}$

16

Review Later



How many different paths are there from A to *any spot* on the bottom row if a bee can only go from one hexagon to a neighboring hexagon that is one of the two below it?

17

Review Later

What is $\frac{5}{14} + \frac{4}{77} - \frac{5}{66} + \frac{7}{78} + \frac{1}{182}$?

Your answer should be a fraction in simplest form, without spaces (e.g. 1/2).

Super Bonus Questions

These questions are an extreme challenge! Some of the most interesting, creative and clever questions are the hardest ones. So try every single one. Even if you can't figure it out, you'll still be deepening your understanding of the important math concepts that underlie this course.

18

Review Later

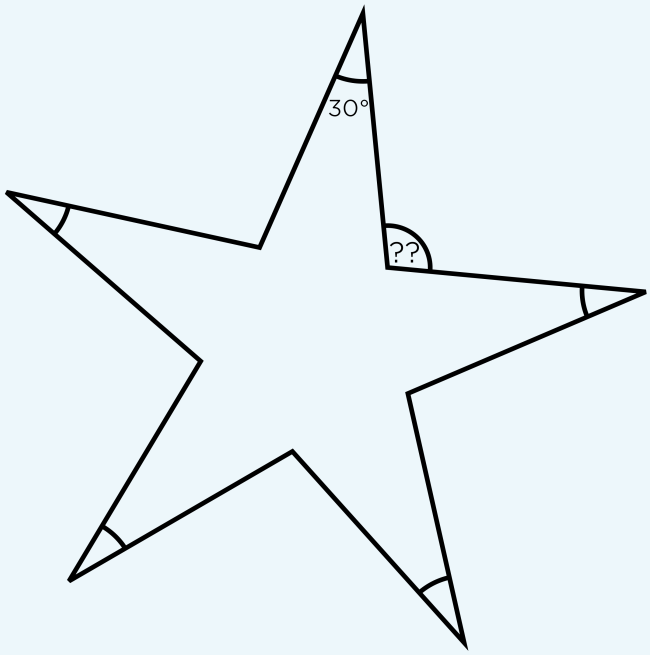
Which of the following is equivalent to

$$\frac{3\sqrt{3}-2\sqrt{2}}{6\sqrt{6}}+\frac{8-3\sqrt{3}}{24\sqrt{3}}+\frac{5\sqrt{5}-8}{40\sqrt{5}}+\frac{6\sqrt{6}-5\sqrt{5}}{30\sqrt{30}}?$$

- $\frac{3\sqrt{3}-1}{6\sqrt{6}}$
- $\frac{1-3\sqrt{3}}{6\sqrt{6}}$
- $\frac{\sqrt{3}-1}{2\sqrt{6}}$
- $\frac{1-\sqrt{3}}{2\sqrt{6}}$

19

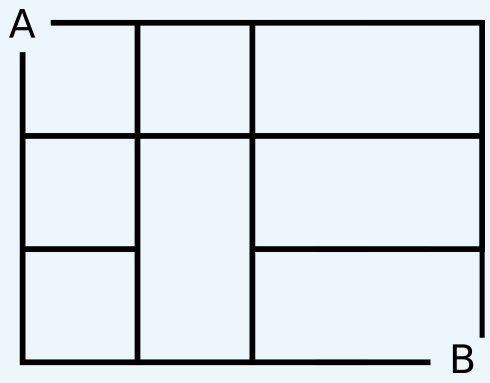
Review Later



How big is the marked angle, in degrees?

20

Review Later



How many different paths are there from A to B if you can only go down and to the right?

View Marked Questions

Finished

Software by
brillium
Version 11.2

Privacy Policy. Assessment content is copyright 2026, Expii, Inc..